EDF optimality wrt minimizing max lateness: proof

Proof.

- Consider the transposition by Dertouzos (see EDF optimality proof)
- A transposition between two slices Σ_a and Σ_b cannot increase $L_{\max}$
  - Let Σ_a be anticipated (i.e., $f'_a < f_a$) and Σ_b be postponed (i.e., $f'_b > f_b$)
  - If $L'_a \geq L'_b \Rightarrow L'_{\max} = L'_a = f'_a - d_a < f_a - d_a = L_{\max}$ given that $f'_a < f_a$
  - If $L'_a \leq L'_b \Rightarrow L'_{\max} = L'_b = f'_b - d_b = f_a - d_b < f_a - d_a = L_{\max}$ given that $d_a < d_b$
- By a finite number of transpositions, σ can be transformed in σ_{EDF}, and, given that the maximum lateness cannot increase, σ_{EDF} is optimal

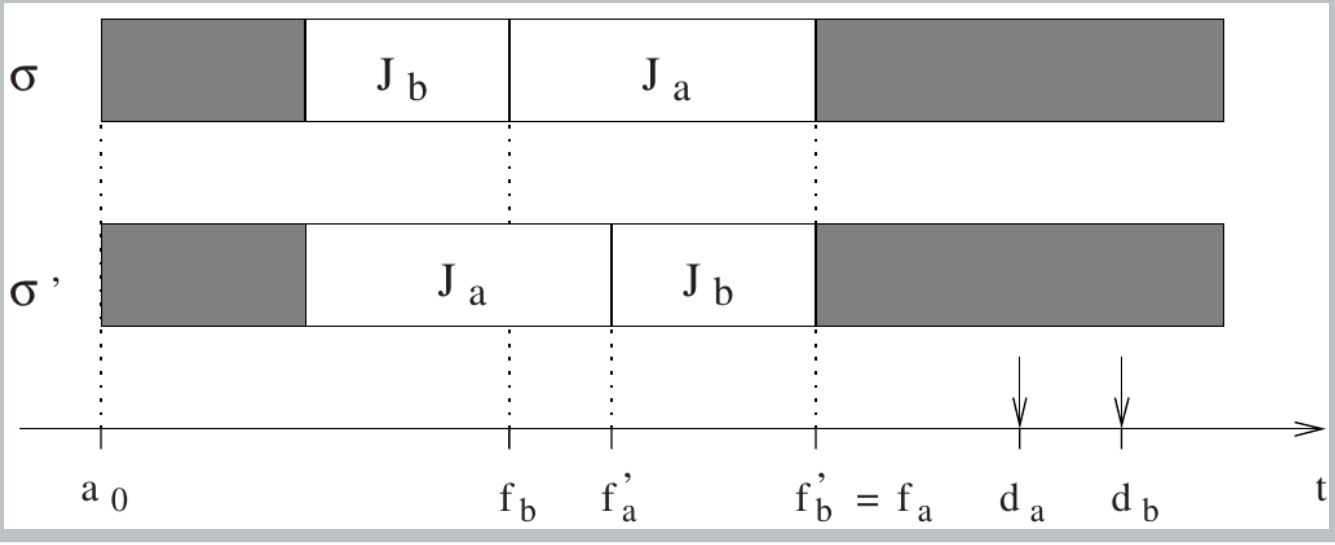

Image from "Hard real-time computing systems" by Prof. G. Buttazzo